Agencia
Nacional de
Investigación
y Desarrollo

Ministerio de Ciencia,
Tecnología, Conocimiento
e Innovación

Gobierno de Chile

Proyecto Puelche: Agent-Based AI for Public Procurement Evaluation

Concurso Investigación Tecnológica 2026

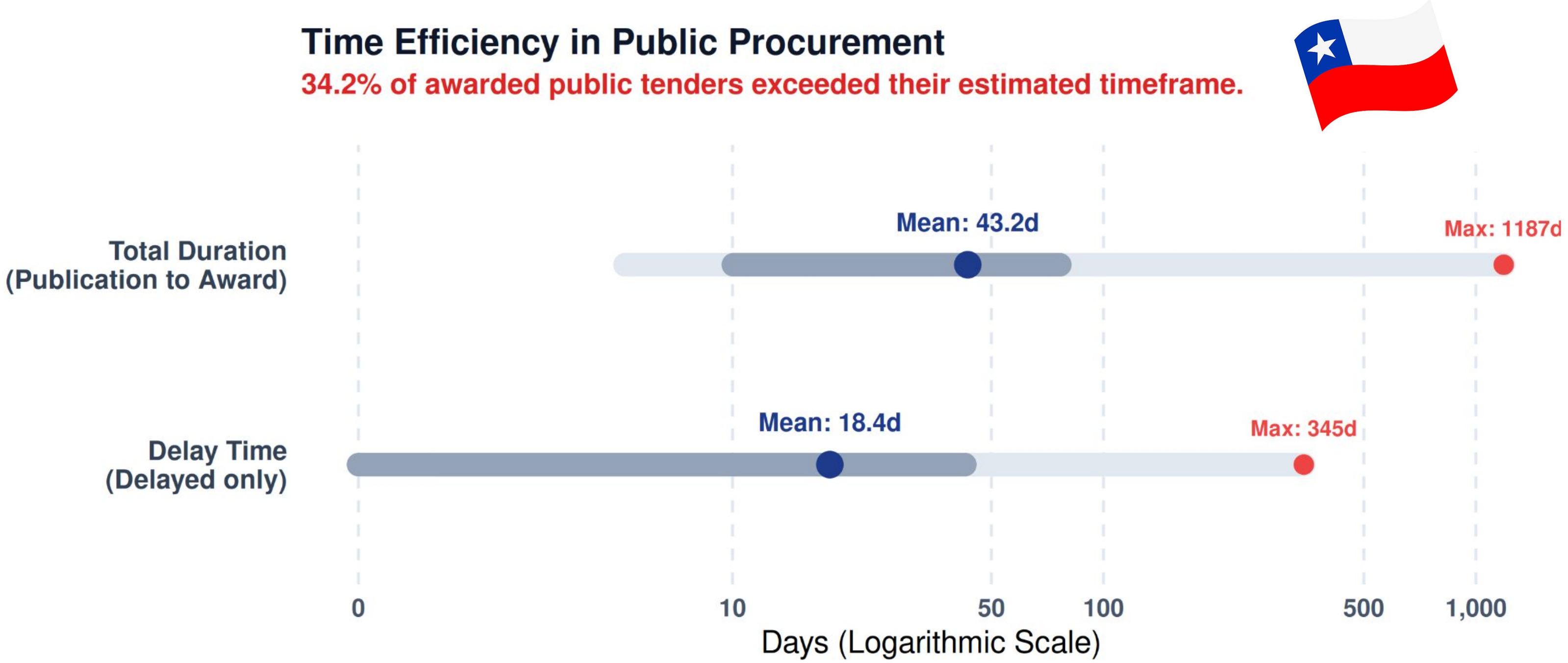
Chapter 1

Description of the problem or opportunity and solution

Chapter 1: Problem/opportunity:

AI Opportunity in Public Procurement

- Public procurement is a strategic policy instrument, representing ~12.7% of GDP and ~30% of government expenditure in OECD countries.
- Systems face significant inefficiencies: high administrative burden, transaction costs, and frequent procedural delays.
- Tender evaluation is highly labor-intensive, requiring extensive manual review and up to 140 person-hours per process.
- Delays affect ~50% of processes, increasing costs, reducing efficiency, and limiting market participation.
- Administrative capacity—not funding—is critical to reducing delays and tender cancellations.
- Current monitoring of procurement efficiency is insufficient; only ~45% of jurisdictions systematically measure performance.
- Regulatory reforms aim to streamline processes but alone are insufficient to address operational complexity.
- Existing AI applications focus on administrative automation, not substantive evaluation of complex tenders.

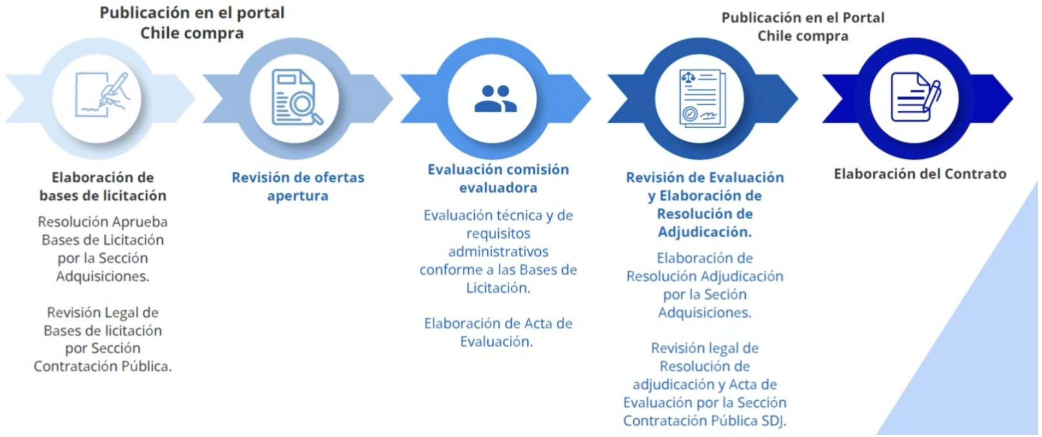


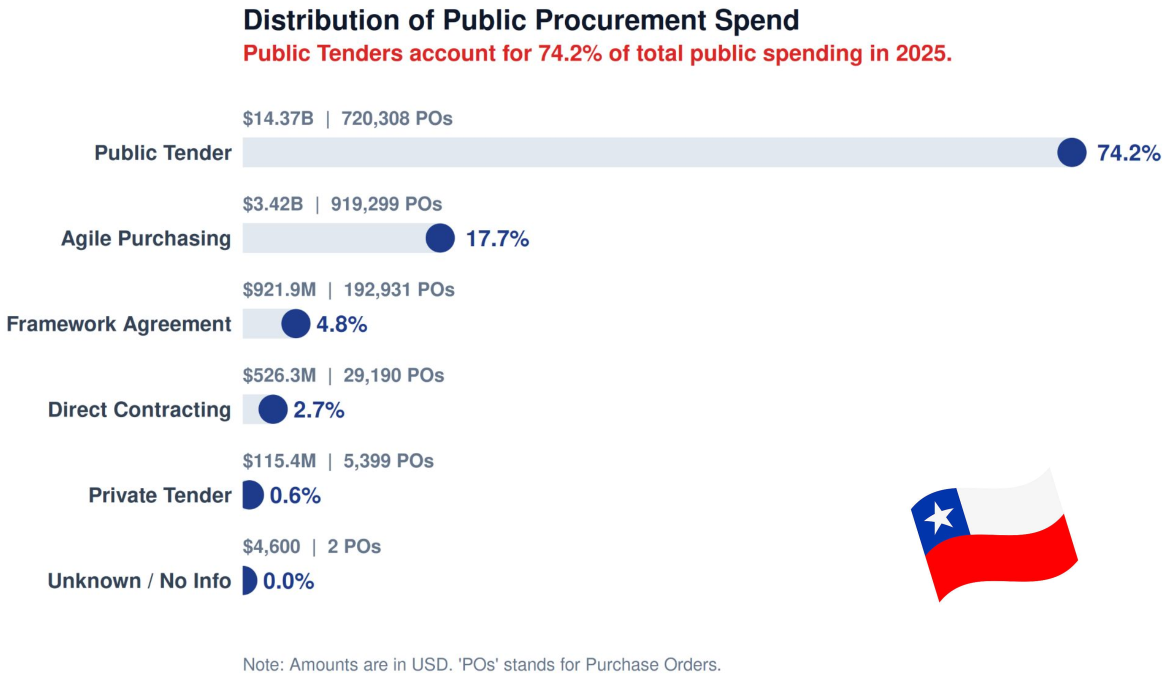
Note: Light gray bar indicates total range (Min-Max). Dark gray bar indicates ±1 Standard Deviation.

Chapter 1: Problem/opportunity:

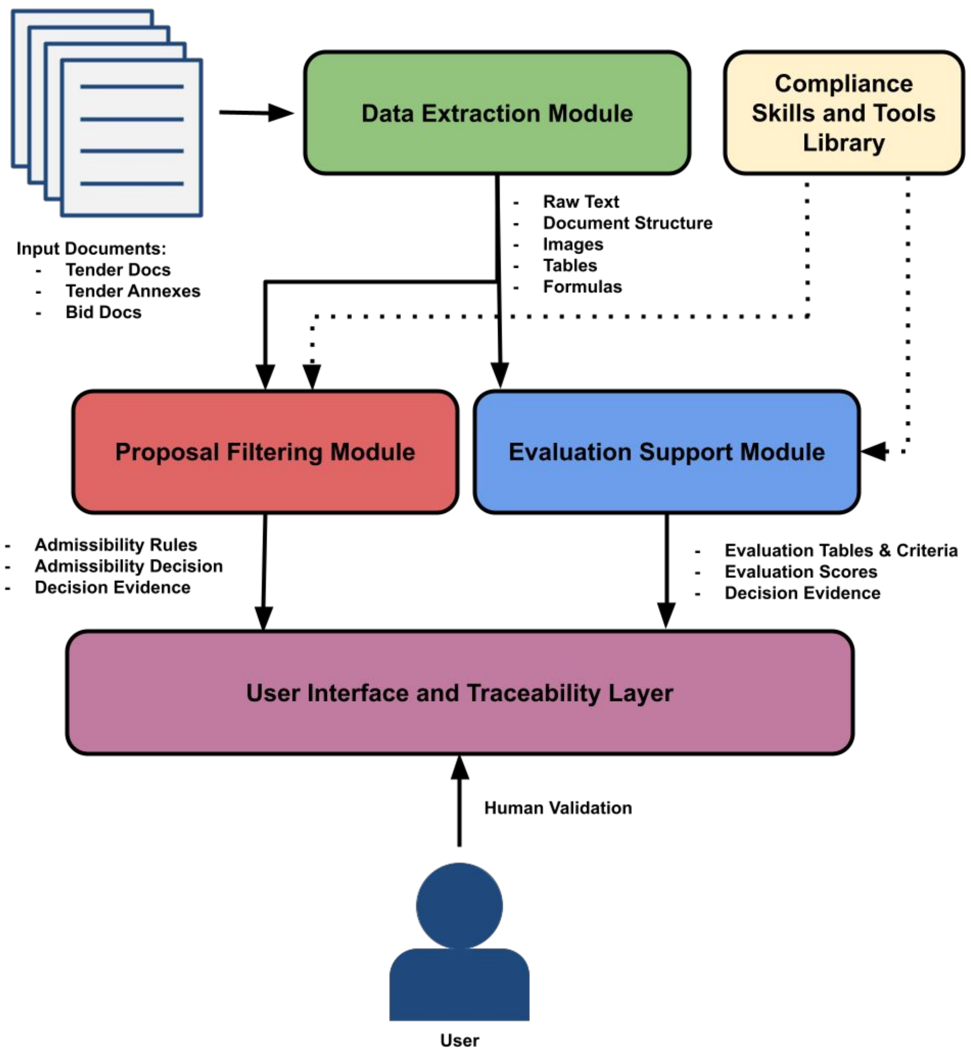
Public Procurement Bottleneck in Chile

- Public procurement represents 14.3% of government expenditure, with US$17.6B transacted via Mercado Público (2024).
- Public tenders dominate procurement (74.2% of spend), forming the core but most complex and resource-intensive mechanism.
- Evaluation phase is the critical bottleneck: admissibility filtering and technical-economic scoring require intensive manual review.
- Evaluation determines fund allocation, ensures compliance, and is the primary safeguard against corruption and inefficiency.
- 62% of procurement lawsuits (2010–2024) originate from evaluation and award disputes.
- Average procurement duration: 43.2 days; 34.2% exceed deadlines, with delays averaging 18.4 days.
- Extreme inefficiencies observed: delays up to 345 days; total processes up to 1,187 days.
- High-impact sectors affected: public works (47.5% delays) and municipalities (41.8%).
- Chile has advanced digital infrastructure (Mercado Público, open data, APIs), but evaluation remains manual.
- **Opportunity:** integrate AI-assisted evaluation to reduce delays and transaction costs, improve consistency, and enhance transparency.





Chapter 1: Proposed Solution and Previous Technological Result



Puelche System

- Supports the evaluation of bids submitted to Chilean public procurement processes
- Reduces administrative burden, transaction costs, and manual document review
- Preserves transparency, traceability, and compliance with Chilean procurement law
- Decision-support tool and all final decisions remain under human authority

Design

- Modular multi-agent architecture
- Decomposes complex procurement analysis into specialized functional modules

Core Functional Modules

1. Data Extraction
   - Ingests documents and produces structured machine-readable formats
   - Performs OCR, layout parsing, and metadata enrichment
2. Proposal Filtering
   - Verifies admissibility requirements of a bid based on the tender documents.
   - Extracts admissibility rules and produces admissibility scores based on those rules.
3. Evaluation Support
   - Supports verifying proposal information and producing scores during evaluation phase.
   - Identifies and structures evaluation criteria defined in the tender documentation.
   - Plans and orchestrates evidence retrieval and analysis; and produces evaluation scores.
4. Compliance Skills and Tools Library
   - Provides a shared capability layer that supports the agents operating in other modules.
   - Offers modular skills and tools that implement specialized operations needed by agents.
   - Implements interoperable interfaces based on emerging agent capability standards.
   - Reusable technological foundation for future AI-assisted administrative systems.
5. User Interface & Traceability
   - Secure web interface for evaluation committees
   - Enables human validation, annotation, and review
   - Shows full traceability to source documents and audit records
6. Optimization Loop
   - Improves the behavior of the agentic system through an optimization process.
   - Progressively expands the system’s capabilities and reliability.

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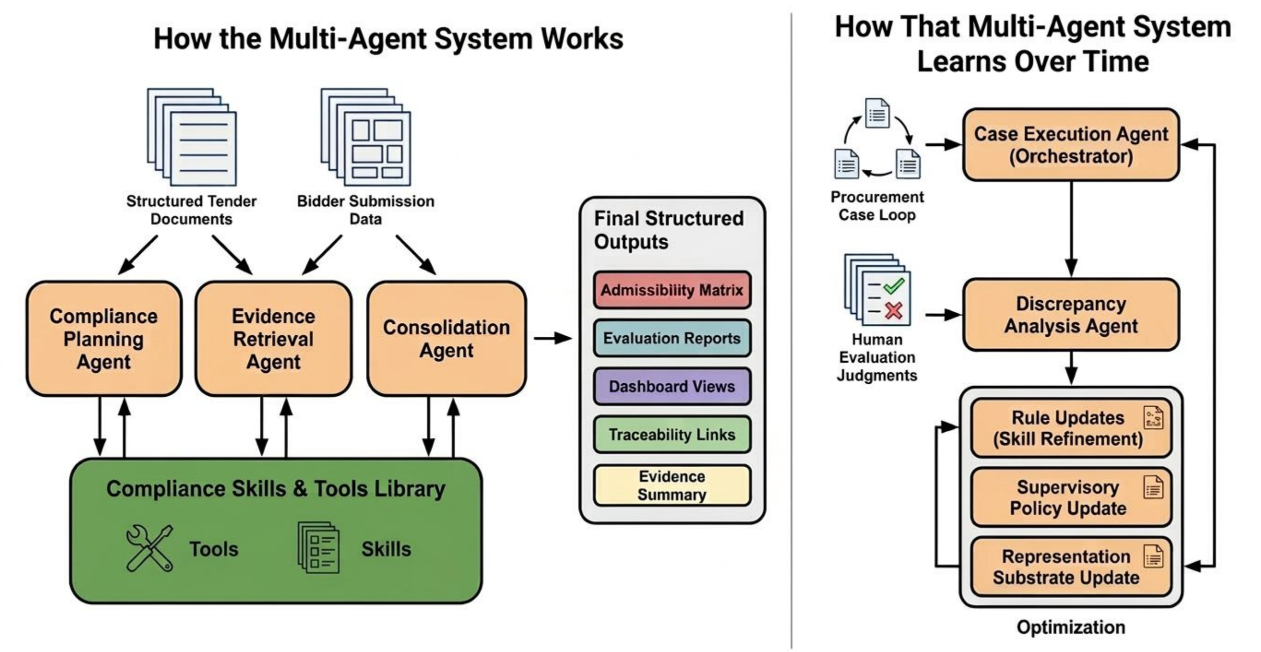


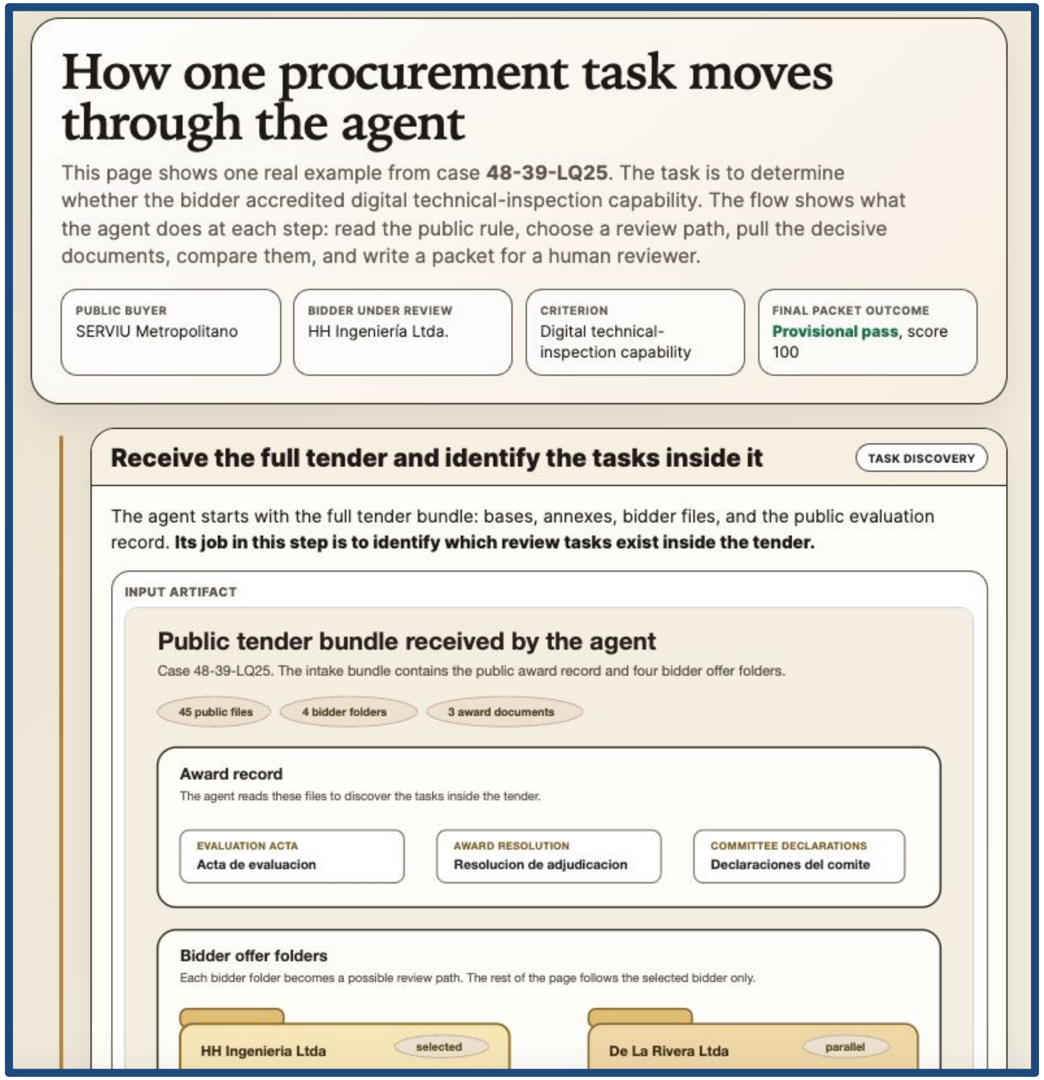
Chapter 1: Proposed Solution and Previous Technological Result

Development Status and Target Maturity

- The current prototype is at **TRL-4**, validating AI-assisted procurement evaluation in a controlled environment.
- The prototype focuses on executive team evaluation, a complex, document-intensive procurement task.
- The system extracts structured evidence from CVs, diplomas, contracts, and supporting documentation.
- Demonstrates feasibility of **agentic architecture for document analysis and evaluation support**.
- The project expands modules: document ingestion, proposal filtering, and broader evaluation dimensions.
- Target **TRL-7 operational prototype** enabling real deployment in SERVIU and public procurement units.

Differentiating Attributes

- Agent-based architecture supporting analytical tasks in public procurement bid evaluation.
- Designed specifically for Chilean procurement procedures and institutional evaluation workflows.
- Uses explicit procurement rules to ensure interpretability and auditability of evaluation logic.
- The outer-loop optimization framework enables continuous improvement from evaluated procurement cases.
- Evolves beyond static prompt-based systems while preserving transparency and traceability.



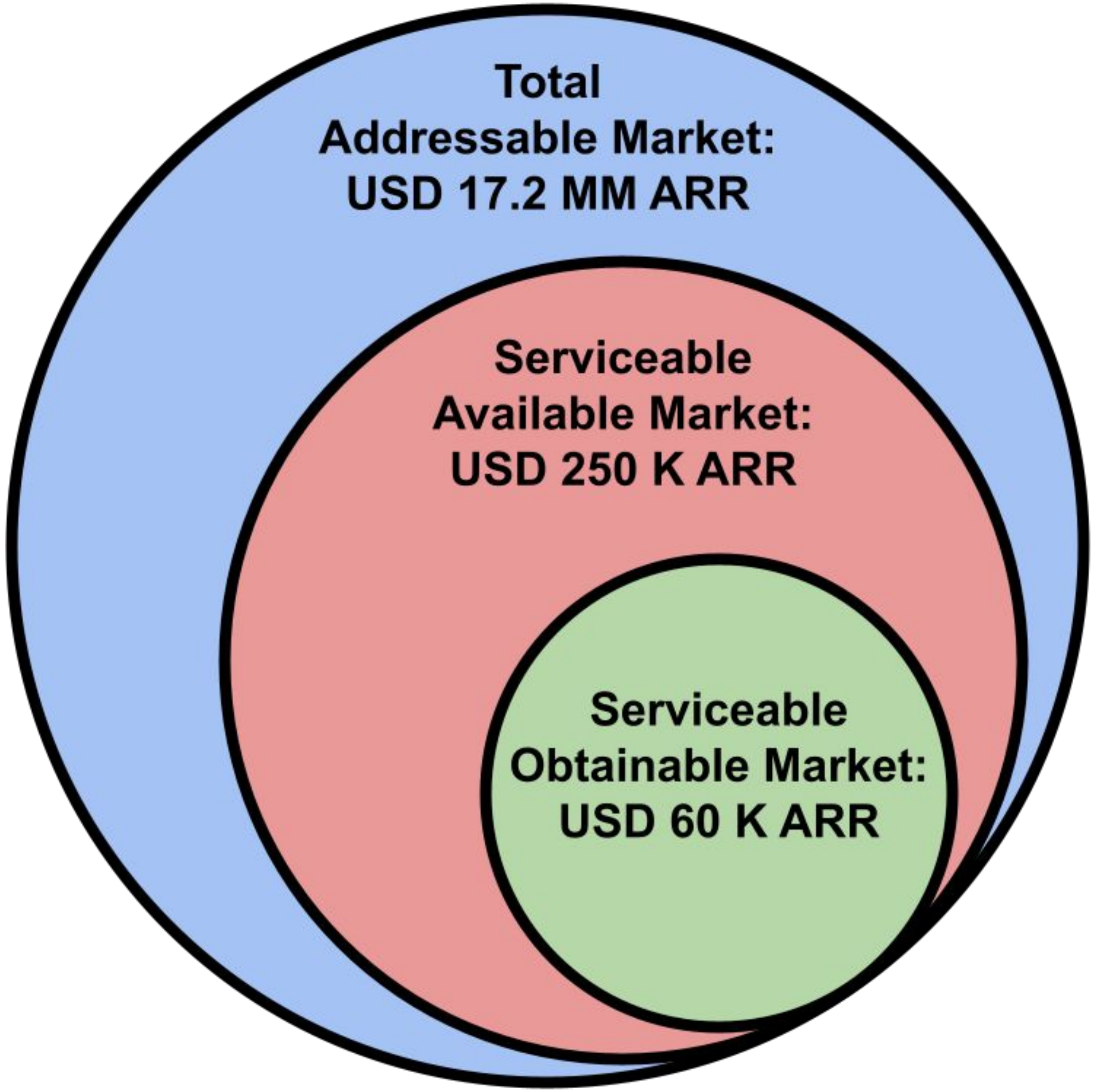
Users and Market Description

Target Users

- **Primary end users:** acquisition units, evaluation committees, and legal/financial teams responsible for verifying admissibility, analyzing proposals, and conducting tender evaluations.
- **Core need:** reduce administrative burden and transaction costs in document-intensive bid evaluation while preserving reliability, transparency, and regulatory compliance.

Market Opportunity

- **Institutional adopters:** public institutions conducting procurement through the Mercado Público platform, currently comprising 1,086 public buying entities across the Chilean government.
- **Market scale:** Chilean public procurement reached USD 17.6B in 2024, with 2.03M purchase orders and 804,000 public-tender orders, highlighting large demand for evaluation support tools.
    - We estimate the TAM for Puelche to correspond to approximately USD 17.2 million in recurring annual revenue.
- **Initial deployment market (SAM):** regional SERVIU institutions of MINVU (16 entities), enabling controlled sectorial validation and early institutional adoption. **USD 253,440 in annual recurring revenue**
- **Short-term obtainable market (SOM):** adoption by four institutions following pilot deployment, approaching operational break-even through recurring institutional subscriptions. **USD 63,360 in recurring annual revenue**.
- **Future expansion: potential** adoption by private suppliers preparing tender proposals and potential international adaptation and integration with the Mercado Público ecosystem, enabling scalable national distribution across public institutions.



Users and Market Description

Key competitive advantage

- **First-mover advantage:** First agent-based AI designed specifically for Chilean public tender bid evaluation support.
- **Interpretability:** Extracts and formalizes procurement rules for interpretable, auditable evaluation processes
- **Operational Efficiency:** Reduces evaluator workload, transaction cost, document analysis burden, and procurement processing times
- **Continuous Improvement:** Outer-loop optimization enables improvement using historical procurement cases and evaluations.

Entry Barriers, Risks, and Externalities

- **Cost–benefit:** AI-assisted analysis reduces manual document review time, lowering administrative costs and enabling faster procurement evaluations while allowing evaluation committees to focus on substantive technical judgment.
- **Operational impact:** Structured evidence extraction and automated compliance verification improve consistency and traceability in evaluation processes, reducing errors and strengthening auditability in public procurement workflows.
- **Regulatory and Institutional Barriers:** Chilean procurement regulations require human authority in evaluation decisions; the system mitigates this through a human-in-the-loop decision-support architecture that preserves evaluator responsibility.
- **Organizational and Adoption Barrier:** Adoption may face institutional caution toward automated tools; mitigated through pilot deployments, shadow-mode testing, and transparent outputs that allow evaluators to verify supporting evidence.
- **Technological Barrier:** Heterogeneous procurement documentation (scanned PDFs, mixed layouts, tables) complicates automation; addressed through OCR, layout-aware parsing, and structured information extraction pipelines.
- **Positive externalities:** improved efficiency in procurement evaluation can accelerate public project execution, strengthen transparency and oversight, and contribute to broader digital modernization of public-sector administrative processes.

Associated Entities and development strategy

SERVIU Región Metropolitana

- Serves as the project’s early adopter, providing a real operational procurement environment for system validation.
- **Institutional mission:** Implement national housing policies locally to promote equitable, decentralized, and participatory urban development, creating sustainable, integrated neighborhoods that improve people’s quality of life and strengthen communities.
- **Operational scale:** Manages high-volume public tenders for housing and urban infrastructure, approximately CLP $62B in 447 purchase orders during the year 2025 under Chile’s procurement framework.

SERVIU
Región Metropolitana
Ministerio de Vivienda y Urbanismo

Collaboration Activities

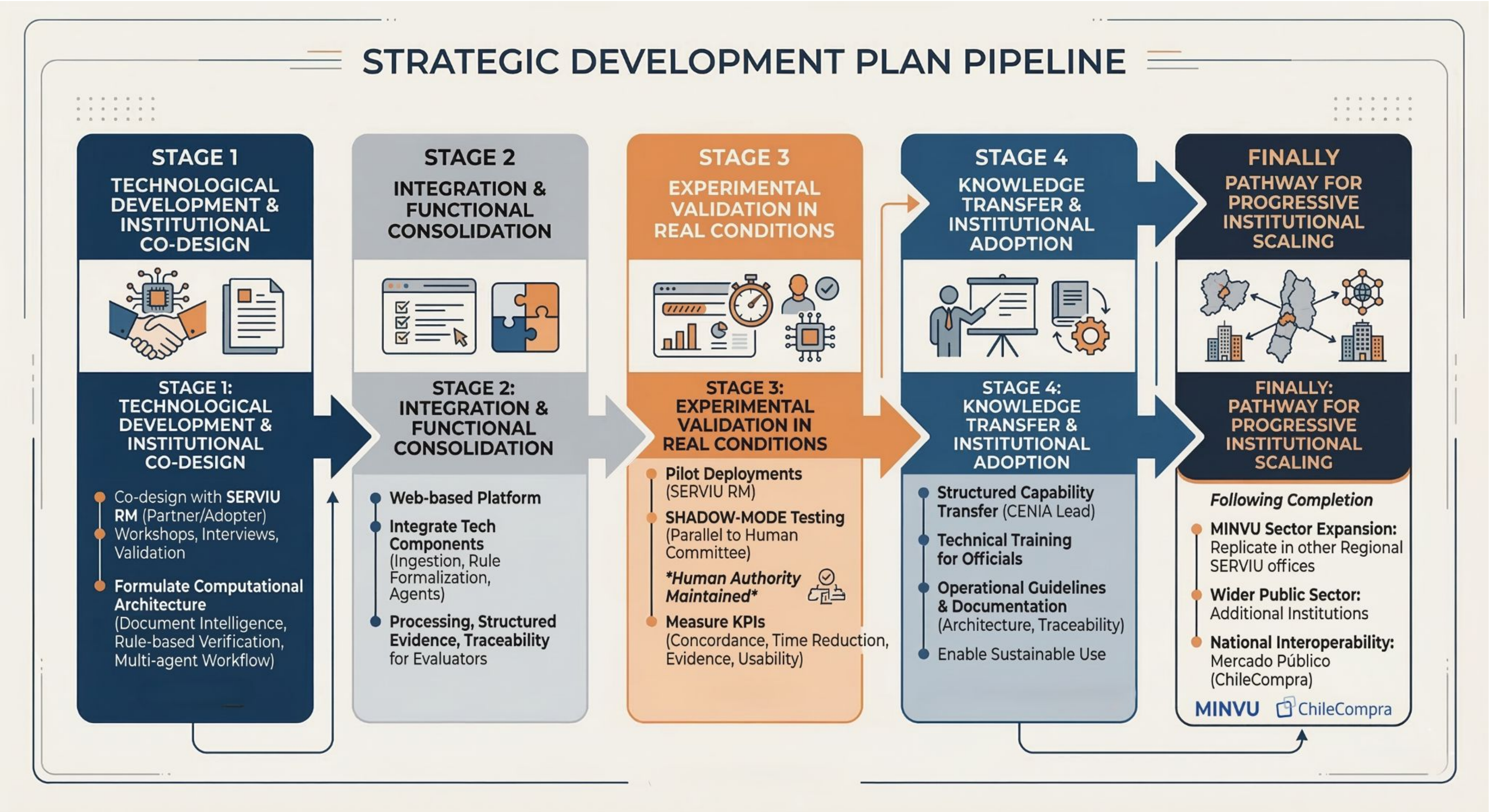
- **Data provision:** Will provide access to historical tender documents, proposals, evaluation reports, and adjudication records for system training and benchmarking.
- **Process alignment and gap identification:** provide continuous domain guidance to map real procurement workflows, clarify implicit evaluation practices, and identify gaps in the system’s understanding of the evaluation process, ensuring alignment with institutional criteria and decision logic.
- **Validation approach:** A shadow-mode validation framework will compare system outputs with human committee evaluations without affecting official procurement decisions.
- **Evaluation metrics:** Performance indicators include admissibility detection accuracy, inter-rater reliability with historical evaluation report gold-standards, processing time reduction, transaction cost impact, and user/institutional satisfaction.
- **Co-development process:** SERVIU RM teams will participate in qualitative assessments, co-design workshops, validation sessions, and iterative testing cycles during system development.
- **Knowledge transfer:** CENIA will implement a knowledge transfer program including technical workshops, training for procurement officials, and operational documentation.

Associated Entities and development strategy

Scaling, Transfer, and Sustainability Strategy

- **Sectorial scaling pathway:** Initial deployment with SERVIU Región Metropolitana, followed by replication across regional SERVIU offices under MINVU with comparable procurement procedures and evaluation frameworks.
- **National expansion:** Progressive adoption by ministries and agencies conducting document-intensive procurement evaluations (e.g., infrastructure, public works, regional development).
- **Ecosystem integration:** Potential interoperability with Mercado Público through collaboration with ChileCompra, enabling broader institutional diffusion across Chile's e-procurement system.
- **Deployment model:** Secure SaaS platform providing centralized maintenance, regulatory updates, and support for adopting public institutions.
- **Technology transfer:** Institutional service agreements and collaborative implementation projects coordinated by CENIA's technology transfer and innovation units.
- **Revenue / savings model:** Institutional service contracts for procurement analytics support, generating efficiency gains through reduced evaluation time, administrative workload, and document-processing costs.
- **Key stakeholders:** SERVIU RM, MINVU regional SERVIU network, ChileCompra, other ministries and public agencies, and multilateral partners supporting procurement modernization.
- **International diffusion:** Gradual expansion through regional networks (IDB, RICG, OECD) targeting countries with comparable digital procurement systems and regulatory frameworks.
- **Regulatory and adoption risks:** compliance with public procurement, information security, and cybersecurity regulations; institutional adoption barriers; and the need to preserve human decision-making authority in evaluation processes.
- **Technical and operational barriers:** Ensuring cybersecurity, regulatory updates, interoperability with existing systems, and sustainable operation through hybrid cloud infrastructure managed by CENIA.

Development Strategy



CHAPTER 2

OBJECTIVES, EXPERIMENTAL DESIGN, GANTT CHART (WORK PLAN), BUDGET, AND RESOURCES

General and specific objectives

General Objective

Develop an Agentic System capable of making Chilean public procurement processes more efficient, without compromising transparency and trust and complying with current regulations, through automatically detecting inadmissible proposals and providing informed prior scores during the evaluation phase. Ultimately, we will make the state contracting process more efficient, thereby improving its policy implementation capacity.

Specific Objectives

1. Characterize and model the institutional process for evaluating public tenders at SERVIU Metropolitano, identifying workflows, eligibility criteria, and evaluation criteria used in practice.
2. Design and develop a technological architecture based on multi-agent systems capable of supporting key tasks in the evaluation workflow, including proposal filtering and evaluation support.
3. Integrate the technological components developed into a functional platform that supports the work of the evaluation committees through document analysis tools, requirements verification, and evidence organization.
4. Validate the system experimentally in collaboration with SERVIU Metropolitano through pilot tests in real bidding processes under a shadow-mode scheme.
5. Transfer the knowledge generated to the SERVIU Metropolitano team through training and technical support activities that facilitate the adoption of the developed system and strengthen institutional capacities for the use of artificial intelligence in the evaluation of public tenders.

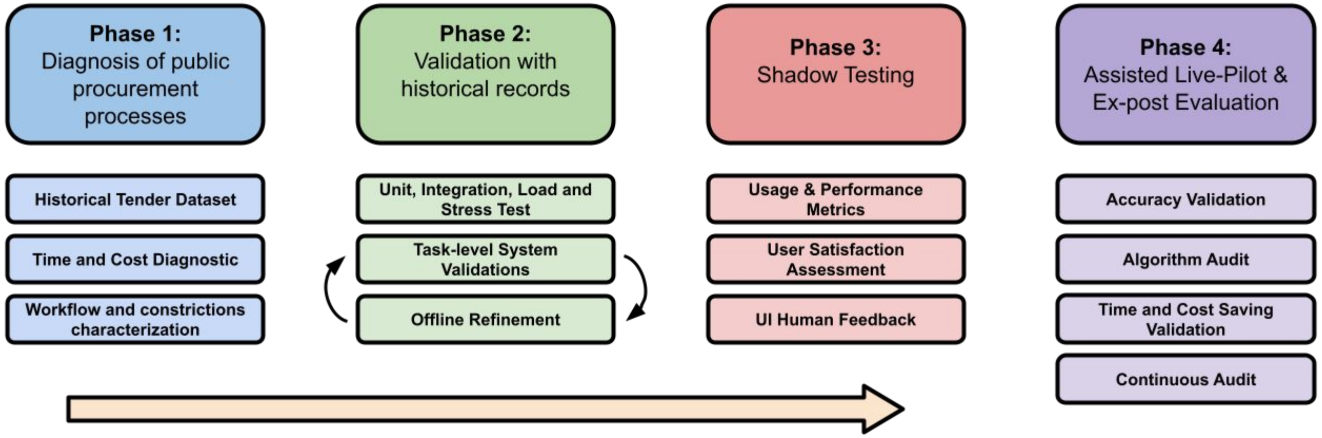
Experimental Design

Phase 1: Ex-Ante Diagnosis & Baseline Setup

- Establish institutional baseline for SERVIU procurement efficiency.
- **Baseline cost/time metrics:**
  - Quantify evaluator workload and estimate transaction costs using methodology by Balaeva et al., PwC.
  - Analyze Mercado Público data to identify timing patterns, delays, tender types, and procurement burden at national and SERVIU RM levels.
  - Build multi-year historical baseline of costs, timing, and workload.
- **Workflow Map:** Map real evaluation workflows and identify high-burden review tasks.
- **Historical Dataset:**
  - Curate historical dataset and define data governance and access protocols for subsequent validation phases.
- Define data access and **governance instruments**.

Phase 2: Controlled Validation & Performance Metrics

- Controlled validation of Puelche prototype using historical procurement cases
- Validate upstream pipeline: document ingestion, OCR, layout parsing, rule extraction, evidence detection, and tool-calling reliability.
- Execute unit, integration, load, and stress tests to ensure system robustness and end-to-end functionality.
- Perform retrospective task-level validation measuring reliability via inter-rater agreement with human evaluations using Cohen’s Kappa, weighted Kappa, and F1 Score.
- Monitor biases, error patterns, OCR quality, token usage, compute cost, and system logs for continuous improvement.
- Advance to shadow mode only upon meeting predefined performance and reliability thresholds.



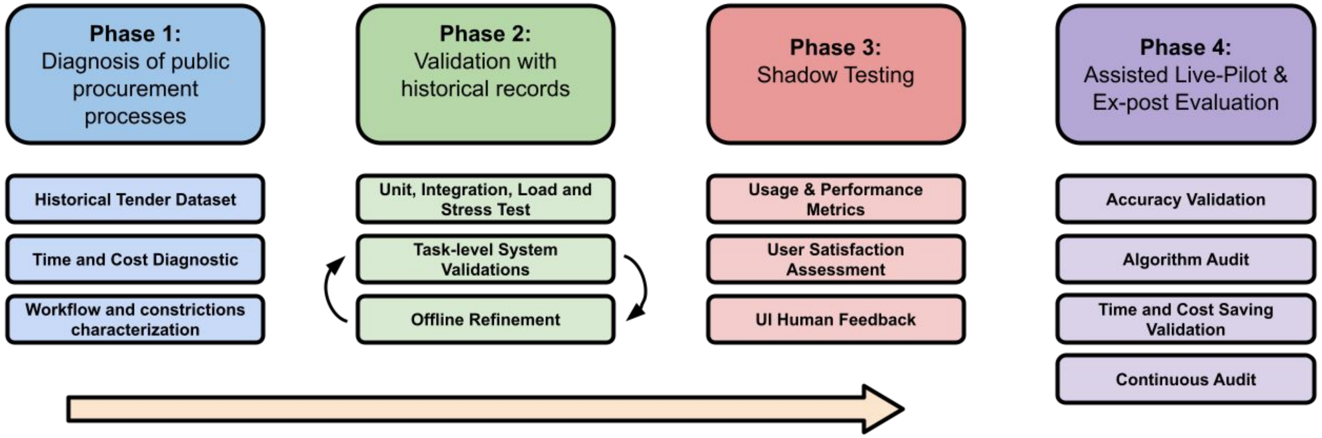
Experimental Design

Phase 3: Shadow-Mode Operational Validation

- Prototype deployed in SERVIU under shadow-mode enabling real-world operational testing.
- System generates task-level packets and case summaries in parallel to official evaluation processes.
- Evaluates not only alignment with prior outcomes, but also usability, interpretability, and operational value for evaluators.
- User evaluation combines observation, focus groups, and surveys assessing usefulness, reliability, transparency, and trust.
- Leadership assessment examines decision quality, procedural consistency, and alignment with institutional objectives.

Phase 4: Live Pilot and Impact Validation

- Transition to assisted live pilot, where committees use Puelche outputs as the starting point for formal procurement decisions.
- Evaluates three core outcomes: accuracy, timeline acceleration, and administrative cost reduction.
- Timeline impact assessed by comparing historical API procurement data with live pilot performance metrics.
- Cost reduction evaluated via baseline vs AI-assisted transaction cost comparison.
- Four-step economic evaluation quantifies net administrative savings per tender.
- Algorithmic audit analyzes mismatched cases to validate bias mitigation and non-discrimination safeguards.
- Establishes continuous monitoring via usage metrics, automated reporting, and in-app feedback mechanisms.



Gantt Chart and Budget

OBJECTIVES	ACTIVITY	ACTIVITY MANAGER (indicate the person from the research team responsible for the execution of each activity)	MILESTONE	"START OF THE ACTIVITY (indicate the month in which the activity will begin)"	ACTIVITY PERIOD (months)	MILESTONE ACHIEVEMENT (indicate the month in which the milestone will be achieved with respect to the total duration of the project)
Objective 1.- Characterize and model the institutional process for evaluating public tenders at SERVIU Metropolitano, identifying workflows, eligibility criteria, and evaluation criteria used in practice.	1. Consolidate existing data (interviews + beta testing): organize findings, map the current workflow, and validate critical points, roles, and process variations with SERVIU.	Director/Principal Investigator/Research Assistant	Milestone 1 - Development and Technical Validation of the Puelche AI-assisted Platform	1	2	10
	2. Close data collection gaps: conduct focused interviews/workshops to capture exceptions, special cases, implicit criteria, and undocumented decisions.	Director/Principal Investigator/Research Assistant		2	2	
	3. Document review and file typology: compile and classify existing tender documents/proposals (types of tenders, typical documents, sections/annexes) and define minimum metadata requirements.	Director/Principal Investigator/Research Assistant		2	2	
	4. Synthesis and prioritization of requirements (effort-impact): consolidate all collected data into a backlog of needs and functionalities, prioritized using an effort-impact matrix, defining the recommended scope for each stage and success criteria.	Director/Principal Investigator/Research Assistant		1	3	
Objective 2.- Design and develop a technological architecture based on multi-agent systems capable of supporting key tasks in the evaluation workflow, including proposal filtering and evaluation support.	1. Define the multi-agent architecture and functional scope: specify sub-agents, responsibilities, available tools, communication/orchestration mechanisms, and input/output contracts for each prioritized use case.	Director/Principal Investigator		2	3	
	2. Design and implement the data extraction module: file loading and validation, content extraction (text/OCR if applicable), segmentation by sections/attachments, and normalization to prepare documents for analysis.	Director/Principal Investigator		3	3	
	3. Build the enabling components as reusable services: proposal filtering module, evaluation support module, data schema and storage, indexing/search, and evidence repository with source references (page/section).	Director/Principal Investigator		4	6	
	4. End-to-end technical integration of the multi-agent MVP: Connect pipeline, services, and agents in a complete flow, incorporating traceability (logs/audits), error handling, and configuration/versioning of criteria to iterate on the TRL 4 prototype.	Director/Principal Investigator		7	7	
Objective 3.- Integrate the technological components developed into a functional platform that supports the work of the evaluation committees through document analysis tools, requirements verification, and evidence organization.	1. Iterate the beta web platform (UX + roles): improve flows and screens for each role (evaluator/supervisor) and incorporate a traceability/explainability layer to understand "why" the system suggests certain things.	Development Team		5	6	
	2. Strengthen the data extraction module (end-to-end PDF): Enhance PDF upload/validation, OCR where applicable, segmentation by sections/attachments, and a viewer that allows navigation of pages and versions within the document.	Development Team		6	6	
	3. Implement requirements verification and evidence organization: automated checklists, alerts for missing/inconsistencies, and an evidence repository by criterion with clear references (page/section) for auditing.	Development Team		7	6	
	4. Full component integration and platform operation: connect the document pipeline, services (rules/evidence), agents, and UI, and include feedback mechanisms in the interface (e.g., categorized warning buttons and a comment field) to capture improvements and errors.	Development Team		10	4	
Objective 4.- Validate the system experimentally in collaboration with SERVIU Metropolitano through pilot tests in real bidding processes under a shadow-mode scheme.	1. Design the validation protocol and SERVIU baseline: select pilot tenders, define what will be compared against the current process (admissibility and scores), and how time/workload and evidence quality will be measured.	Development Team	Milestone 2 - Institutional Transfer and Pilot Deployment of the Puelche Platform to SERVIU	11	2	22
	2. Enable pilot operation and integration with the SERVIU environment: agree on file upload/download flows, profiles/roles, permissions, and a results log, so that the system can function as support without interfering with the official decision.	Development Team		13	6	
	3. Run shadow testing in real-world processes: Run the tool alongside the committee in current tenders, recording times, applied criteria, suggested scores, and associated evidence, plus committee observations/issues.	Development Team		14	6	
	4. Compare results against human evaluation and finalize with an improvement/scaling plan: Analyze differences in admissibility and scores, identify gaps (rules, agents, UI, evidence), and compile a report with recommendations and a roadmap for gradual deployment.	Development Team		19	2	
Objective 5.- Transfer the knowledge generated to the SERVIU Metropolitano team through training and technical support activities that facilitate the adoption of the developed system and strengthen institutional capacities for the use of AI in the evaluation of public tenders.	1. Usability Guide / User Manual: Develop a clear (step-by-step) manual with screenshots, sample cases, and best practices for operating the system in tender evaluation.	Co-director/Development Team		10	5	
	2. Operational Enablement during the Project: Conduct induction sessions and provide limited support during pilots/validation, recording frequently asked questions and usage adjustments to improve the guide.	Co-director/Development Team		12	11	
	3. Technical and operational documentation: Deliver documentation on architecture, basic operation, configuration of rules/rubrics, versioning criteria, and data governance guidelines for internal continuity.	Co-director/Development Team		16	7	
	4. Scalability plan + continuity proposal: Consolidate the backlog of identified improvements (impact-effort), define a roadmap, and structure a commercial proposal for support/maintenance and post-FONDEF evolution (service levels and costs).	Co-director/Development Team		18	7	

Gantt Chart and Budget

- Budget summary table

ITEM	SUBSIDIO SOLICITADO	APORTE(S) BENEFICIARIA(S)	APORTE(S) ASOCIADA(S)	TOTAL
GASTOS EN PERSONAL	137,829,000	43,200,000	47,410,635	228,439,635
EQUIPAMIENTO	0	0	0	0
INFRAESTRUCTURA Y MOBILIARIO	0	17,280,000	0	17,280,000
GASTOS DE OPERACIÓN (considera gastos en los ítems Gastos generales-subcontratos-pasaje internacional-viático internacional- gastos de apoyo a la administración)	64,000,000	0	23,700,000	87,700,000
GASTOS DE ADM. SUPERIOR 15% Máx.	35,171,000	0	0	35,171,000
TOTAL	**237,000,000**	**60,480,000**	**71,110,635**	**372,348,000**
PORCENTAJE	64.30%	16.41%	19.29%	100%

Professional Capacities

Partner Entity: SERVIU Región Metropolitana

Director: Felipe del Río

- Industrial engineer and PhD in Artificial Intelligence (2025).
- Research focuses on the generalization properties of deep learning models.
- Worked in AI since 2017, and prior to his research career worked as a software developer.
- Ample experience in academic and applied contexts.

Deputy Director: Rodrigo Durán

- Commercial Engineer
- Current CENIA’s CEO and previous experience in the science ministry.
- Strong understanding of administrative processes and procurement practices
- Proven leadership in leading multidisciplinary teams and complex projects
- Ability to drive institutional collaboration for effective system development and adoption

Principal Investigator: Mitchell Bosley

- PhD in Political Science from the University of Michigan (2024)
- Postdoctoral Researcher at CENIA conducting research at the intersection of Social Science and Artificial Intelligence. Has published research at Q1 journals including the American Political Science Review (APSR) and the Proceedings of the National Academy of Sciences (PNAS) Nexus.
- Brings experience in designing and testing agentic systems for document intelligence

Research Assistant: Ismael Aguayo

- BSc in Sociology (2025) with distinction. Experience as a research assistant in Millennium Nucleus on Digital Inequalities and Opportunities (NUDOS) and CENIA.
- Strong quantitative, qualitative, and NLP methodological skills.
- Focus on the impact and applications of AI systems in public institutions.

Gender Equity in Project Team Formation

- The project explicitly commits to incorporating female researchers and students.
- Female participation will be promoted through thesis supervision.
- Gender diversity will also be ensured in project activities.

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